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Creating a Crude Oil Inventory Management Program (COIMP) Using MS Excel & AI

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In a Crude Oil Terminal, there is an inflow of different crude oils of various grades having different parameters like API, density, BS&W and free water. When crude oil is received in a tank, the final composition of that tank is derived based on parameters of opening inventory and the parameters of crude oil received. After this, the tank is given settling time for certain hours before mixer operation for excess water draining. Post mixer operation, again settling is given for certain hours for sludge to settle down and then finally, tank is lined up for pumping via pipeline transfer (PLT) to the Refinery. PLT may have a multi stream blender. In our case, a two-stream blender is considered. Apart from this, inter tank crude oil transfer and sea water intake for SPM hose flushing is also done as per requirement. So, basically, the crude oil flow can be primarily divided into 4 categories (Refer Fig. 1):

1. From Motor Tanker to Oil Terminal via SPM or Jetty,
2. From Oil Terminal to Refinery via PLT,
3. Inter tank transfers for diluting crude grades and
4. Water flushing of SPM hoses.

After each of the above crude oil movements, the grade and ratio in a particular tank changes based on the opening inventory in that tank and the quantity of crude oil moved. This change is also reflected in density & BS&W parameters of that tank. So, one task of the **COIMP** is to keep track of crude oil movements in order to capture correct grades, ratios and other parameters for each tank. And the second task is to reconcile all this data for presenting in a systematically organized report for daily, weekly or monthly reconciliation.

Introduction: Basics of Crude Oil Inventory Management Program (COIMP)

In a Crude Oil Terminal, there is an inflow of different crude oils of various grades having different parameters like API, density, BS&W and free water. When crude oil is received in a tank, the final composition of that tank is derived based on parameters of opening inventory and the parameters of crude oil received. After this, the tank is given settling time for certain hours before mixer operation for excess water draining. Post mixer operation, again settling is given for certain hours for sludge to settle down and then finally, tank is lined up for pumping via pipeline transfer (PLT) to the Refinery. PLT may have a multi stream blender. In our case, a two-stream blender is considered. Apart from this, inter tank crude oil transfer

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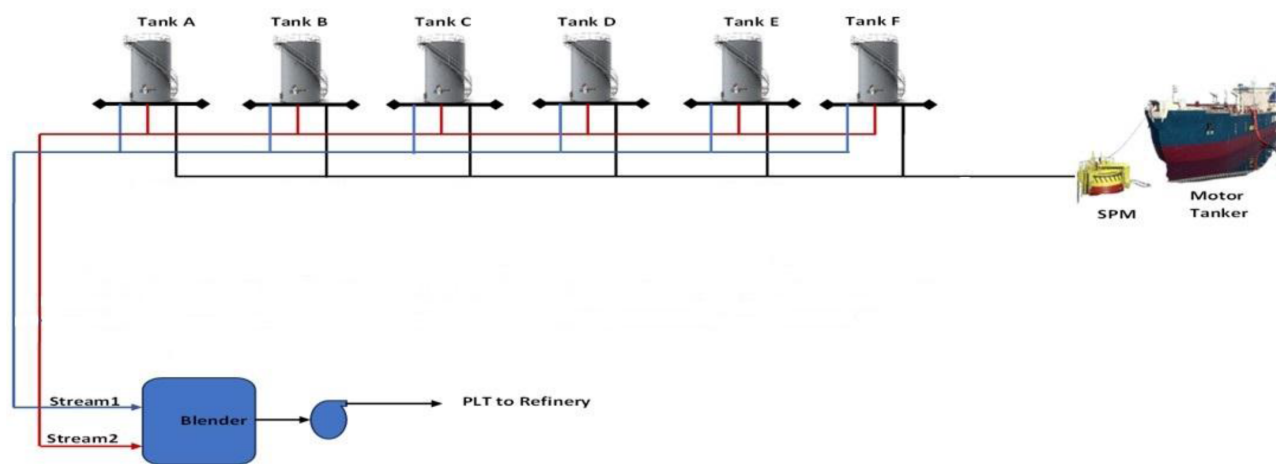


Figure 1—Basic structure of a Crude Oil Terminal of a Refinery.

Detailed structure of 6 Modules of Crude Oil Inventory Management Program COIMP

1st Module – Database to define Crude inventory

The very first Module of **COIMP** is the database of all the parameters with which the crude oil inventory in a terminal is defined. Tank dips, water cuts, density, temperature and BS&W. All this data is captured at midnight every day and stored in a systematic array. Data can be entered manually or auto fetched using softwares like Exaquantum or IP21, etc.

The state of a particular crude tank is defined in this Module. State of a crude tank can be any combination of the following:

1. Receiving
2. Dispatch
3. ITT Source
4. ITT Dump
5. Stand-By
6. Maintenance
7. Dispatch / Receiving – Tank was dispatching crude to Refinery through PLT (Pipeline Transfer). Tank changeover done and handed over for receiving crude from Motor Tanker via SPM on the same day.

8. Receiving Dispatch – Tank was receiving crude from Motor Tanker via SPM. Tank changeover done and handed over for dispatching crude to Refinery through PLT (Pipeline Transfer) on the same day.

In Fig. 2, refer to columns H' and 'N', where the tank states are defined by selecting from a dropdown list for each of the six tanks. Tank States at #7 and #8 have a different array that captures the intermediate parameters while switching from Dispatch to Receiving and Vice-a-Versa. For example, in a day, if one tank was in dispatch and after completing dispatch it was handed over for receiving, then the dip after dispatch and before receiving is recorded in the bottom table of Fig. 2. The Tank states defined in the 1st Module are used in the other Modules in various calculations.

Another important parameter that is captured in this Module is the cut off level in case of PLT batch change. PLT to Refinery is done using a two-stream Blender. One stream takes input from one crude tank and the second stream takes input from a different crude tank. So, at a time two tanks can be blended in a desired ratio. In such a case, when only one stream tank is changed, then the common tank dip is required to calculate the PLT quantities for both the batches. This common tank dip is also recorded in this Module. For example, PLT Batch changed from Tanks 1A:1B to 1B:1C, where tank B is common.

1860 cells are formulated to select or calculate data pertaining to Tank States #5 and #6 for 6 tanks for 31 days.

2nd Module – Database to update Crude Grades and Ratios for each Crude Tank after crude receipt

Crude inventory in a tank is defined as the ratio of different types of crudes present in that tank. It is simply the percentage of each grade present in the total inventory present in that tank. A simple representation of crude grade ratio looks like this- **ARL:ARH:URL:SKF – 20:30:5:45**.

This ratio of crude in a tank keeps on changing with every new crude grade top up (receiving). For example: Let us consider a crude oil tank having an inventory of 10000 KL with crude grade ratio **ARL:ARH:URL:SKF – 20:30:5:45**. This means it has ARL – 2000 KL, ARH – 3000 KL, URL – 500 KL and SKF – 4500 KL. Now, let us say that a top up of 20000 KL URL is done in this tank. The total inventory now becomes 30000 KL and crude ratio is updated as –

$$\text{ARL} = 2000/30000 * 100 = 6.67\%$$

$$\text{ARH} = 3000/30000 * 100 = 10\%$$

$$\text{URL} = 25000/30000 * 100 = 83.3\%$$

$$\text{SKF} = 4500/30000 * 100 = 15\%$$

Now writing crude ratio in decimals does not look good. So, let us approximate decimals into whole numbers. Then, the updated crude grade will be - **ARL:ARH:URL:SKF – 7:10:83:15**. Now, the point to observe here is that to some extent approximation is required. Manually, we can check the sum of percentage is 100% but when we use MS excel for this task, sometimes the percentage sum may be 100 ± 1 due to approximations after decimal. This error will be carried on to the next calculation and so on. Hence, a *hundred % check block* logic is created in MS Excel. This logic is used to adjust the total percentage sum of crude grades to be exact 100%, only if it is otherwise.

One more case of change in crude ratio happens when crude oil from one tank is migrated to another by gravity. This case is called the Inter Tank Transfer (ITT). This is done to mix available grades to get the desired quality of Crude. Let us consider that ITT is done from Tank A to Tank B, having Crude Ratio - **ARL:ARH:URL:SKF – 20:30:5:45** and **BHY:URL:SKF:UPP – 20:30:5:45**, respectively.

A	H	I	J	K	L	M	N	O	P	Q	R	S	
1	Date	01 August 2025					02 August 2025						
2	Tank	R/D/S	Tank Dip as per ATG(CM)	Temp.(Deg C)	BS&W(%)	Density @ 15 °C (gm/cc)	Common Tank Cut Off	R/D/S	Tank Dip as per ATG(CM)	Temp. (Deg C)	BS&W(%)	Density @ 15 °C (gm/cc)	Common Tank Cut Off
3			cms.	oC	(%)	gm/cc			cms.	oC	(%)	gm/cc	
4	1A	D	860.1	31.9	0.05	0.8573	1089.8	D	401.6	31.7	0.05	0.8573	
5	1B	S	1249.4	31.8	0.121	0.8588		S	1249.4	31.7	0.121	0.8588	
6	1C	M			-	-		M			-	-	
7	1D	S	1582.1	33.4	0.05	0.8532		S	1582.5	33.4	0.05	0.8532	
8	1E	D	1471.8	27.3	0.1	0.8689		D	1180.5	28.4	0.1	0.8689	
9	1F	S	1475.1	30.6	0.025	0.8429		S	1475.2	30.6	0.025	0.8429	
16			10792.0	338.4					11085.8	344			
17													
26													
31	Receive /Dispatch cutoff data (Delete data if R/D not selected)						Receive /Dispatch cutoff data (Delete data if R/D not selected)						
32	1A					0							0
33	1B					0							0
34	1C					0							0
35	1D					0							0
36	1E					0							0
37	1F					0							0

Figure 2—A screenshot of Module #1 of COIMP.

Opening		Closing		CrudeTopup Note: First column last crude. Second column new crude and			
Tank	Grade	Ratio	Grade	Ratio	Tank	OpeningQuantity	ClosingQuantity(KL)
1A	ARL:URL	85:15	ARL:URL:ALS	29:39:32	1A	19784.348	57696
1B	BHY:URL:ARL:MUR	1:80:5:14	URL:ARL:MUR	24:72:4	1B	9630.105	31685
1C	0	0	-	-	1C		0
1D	ARL:SKF:URL	26:33:41	ARL:SKF:URL	76:11:13	1D	18894.970	57686
1E	ARL:MUR:SKF:URL	19:5:1:75	ARL:MUR:SKF:URL:ALS	14:3:1:44:38	1E	34027.405	57775
1F	ARL:MUR:URL	15:84:1	ARL:MUR:URL	15:84:1	1F		0

Figure 3—A screenshot of a part of Module #2 of COIMP. Shows crude receipt in Tanks 1A, 1B, 1D and 1E.

=ROUND(IF(A3=025,IF(W13="D",T13,IF(OR(W13="R/D",W13="D/R"),T20,0))+IF(W14="D",T14,IF(OR(W14="R/D",W14="D/R"),T21,0)),IF(A3=026,IF(R13=2,IF(W13="D",V13,V20),IF(W13="D",IF(X13>0,U13,T13),IF(X13>0,U13,T20))+IF(R14=2,IF(W14="D",V14,V21),IF(W14="D",IF(X14>0,U14,T14),IF(X14>0,U14,T21))),IF(A3=027,IF(W13="D",V13,IF(OR(W13="R/D",W13="D/R"),V20,0))+IF(W14="D",V14,IF(OR(W14="R/D",W14="D/R"),V21,0))),0))
=ROUND(IF(A3=026,IF(R15=2,IF(W15="D",U15,U22),IF(W15="D",IF(X15>0,V15,T15),IF(X15>0,V22,T22))+IF(R16=2,IF(W16="D",U16,U23),IF(W16="D",IF(X16>0,V16,T16),IF(X16>0,V23,T23))),IF(A3=027,IF(W13="D",U13,U20)+IF(W14="D",U14,U21),0)),0)

Figure 4—Screenshots of complex formulas used in PLT quantity calculation in Module #3 of COIMP.

Crude inventory in Tank A and Tank B is 30000 KL and 10000 KL, respectively. Let us say 20000 KL crude oil is migrated from Tank A to Tank B. In that case, the Crude Ratio in Tank B is updated as –

$$\text{ARL} = (0.2 \times 20000) / 30000 \times 100 = 13.33\%$$

$$\text{ARH} = (0.3 \times 20000) / 30000 \times 100 = 20\%$$

$$\text{URL} = [(0.05 \times 20000) + (0.3 \times 10000)] / 30000 \times 100 = 13.33\%$$

$$\text{SKF} = [(0.45 \times 20000) + (0.05 \times 10000)] / 30000 \times 100 = 31.67\%$$

$$\text{BHY} = (0.2 \times 10000) / 30000 \times 100 = 6.67\%$$

$$\text{UPP} = (0.45 \times 10000) / 30000 \times 100 = 15\%$$

Approximating decimals into whole numbers, the updated crude ratio in Tank B becomes –
ARL:ARH:URL:SKF:BHY:UPP – 13:20:13:32:7:15

In this module, all the above calculations are done by system. Users only need to enter the inventory of a particular tank before receiving (taken from integrated tank calibration charts), then select the crude type (interlinked with a common crude database) and enter the quantity received for each crude grade top up. The final closing grades are calculated automatically. Some assumptions are made in this module based on actual experiences. These assumptions are that a tank can have maximum 10 numbers of crude grades in it. This is used to accommodate each crude grade in one cell for calculation reference. So, 10 cells per tank are blocked for initial grades. Next, let us say apart from the 10 mentioned grades, some new grades are also added in the tank. So, 5 cells are blocked for this purpose. Then for the initial grade break up, we have 10 cells for grades and ratios, each. Then based on the user inputs for opening inventory, grade type and quantity top up, the system updates the output array, which has 15 cells for grades and ratios each. Out of these 15 cells,

as described earlier, 10 cells are for existing grades and ratios each and 5 cells are for new grades and ratios, each. All decimals are rounded off to whole numbers. After this, the sum of all output ratios is checked for 100%. If it is anything but 100%, then another calculation block, which contains 15 cells for grades and ratios, each, will approximate the total to 100%. Two different logics, one for greater than 100% and one for less than 100% are defined. After this, the output grade and ratio are reflected in one cell each. 11415 cells are formulated for 6 tanks considering 15 receipts (Motor Tanker and ITT combined) in a month.

Updated QC parameters like Density and BS&W after each receipt is also updated in this Module only. These parameters are fetched from this Module into Module 5 for daily reports.

3rdModule – Pipeline Transfer (PLT) batch creation

Pipeline transfer (PLT) of crude oil is done using a set of pumps in parallel, all of which take suction from the output of a two-stream blender. Refer Fig. 1— Basic structure of a Crude Oil Terminal of a Refinery. PLT batch creation is an important step of crude oil inventory management. It gives us a systematic sequence of crude outflow from crude oil terminal to refinery. PLT batches are designed by scheduling & planning depending on the production targets. One PLT batch contains information like batch number, batch name, blending ratio, quantity (KL/MT), Crude Grade, Crude Ratio, density, BS&W, Viscosity, etc. Module 3 eases the procedure of calculating all this info with minimum input and effort.

PLT batches are of two types – pumping both streams in a fixed ratio and single stream pumping. So, a PLT batch change can be any of the following:

1. From single stream pumping to two-stream pumping and vice-a-versa. Ex. 1A – 100% to 1A:1B – 80:20.
2. One tank change in two stream pumping. Ex. 1A:1B-80:20 to 1A:1C-80:20.
3. Both Tank Change in two stream pumping. Ex. 1A:1B-80:20 to 1C:1D-2:80.
4. Blending Ratio change of both tanks in two-stream pumping. Ex. 1A:1B to 80:20 to 1A:1B – 50:50.

To capture all these PLT changes accurately, 3 categories of PLT batch changes are defined in this module – Single Batch, Common Tank Batch and Same Batch New Ratio. These 3 categories along with the tank states and cutoff level values defined in Module 1 (Database to define crude inventory) are together used in a complex logical formula to calculate the date wise PLT quantity.

For calculating the crude grade, ratio and viscosity, this module uses complex blocks of calculations, taking inputs from Module 2 and the integrated crude database. The flow of calculation is such that first, the user enters that PLT batch name and blending percentage in designated cells as 1A:1B and 70:30, respectively. Then, the user selects the motor tanker from a drop-down list (which is extended onto Module 3 from Module 2 using a set of formulas) after which that particular tank crude grade and ratio was created. This is done for each tank in a batch.

After this, the system fetches the crude grades and ratios from Module 2 for further calculation. One calculation block pro-rates the crude ratios as per blending ratios entered by the user in designated cells. Another calculation block searches for crude grades from one tank in another and vice-a-versa. This is done so that if one grade is present in both tanks (streams), then its pro-rated percentage in each tank shall be added and if one grade is unique to one tank, then its pro-rated percentage shall be reflected in the final PLT batch. For Example, Tank A has BHY:ARL – 80:20 and Tank B has BHY:URL – 80:20 and these tanks are blended in a 50:50 ratio. Then the pro-rated ratio of tank A and Tank B will be BHY:ARL – 40:10 and BHY:URL – 40:10, respectively. And the PLT batch Crude grade and ratio will become BHY:ARL:URL – 80:10:10. This is the simplest of example. Practically, one tank has up to 5-7 grades of different crudes. So, for two tanks, one can imagine the complexity of calculations involved.

After this, a *hundred % check block* checks the total sum of all crude percentages in the final PLT batch for 100 total. If the is less than or more than 100, then after final adjustments, the final output of PLT Crude Grade and Ratio is reflected in the designated output cells.

Viscosity calculator block simply uses the final PLT batch crude ratios and respective crude viscosity (fetched from the integrated crude database) to calculate the viscosity in the designated output cell.

So, to summarize, the user only needs to first select the type of PLT batch change from the drop-down list, enter the date, enter batch name and blending ratios, then system itself calculates the PLT quantity, crude grade, crude ratio and viscosity. 250+ complex formulas are used in this module.

Select Batch Type	Date	Qty. KL	Tank Name	Blending Ratio %	Grade	Ratio	Viscosity
Single Batch	2 August 2025	27218	1A:1E	60:40	ARL:ALS:URL	55:5:40	11.0
Single Batch		0	1C:1A	65:35			
Common Tank Batch							
Same Batch New Ratio							
Select Opening/Closing Grade and vessel from dropdown list							
	1A	Closing	MT CENTURY				
	1E	Closing	MT CENTURY				

Figure 5—A screenshot of Module #3 of COIMP. User inputs are colored.

4thModule – Database of PLT batches

In Module (#3), PLT batches are created without any history. In other terms, when a new batch is created, the old one gets overwritten. Module #4 is designed to store all PLT batches in one place. This database is used to fetch data for generating periodic reports. To start with, a PLT batch list is needed as a base. For example, at the beginning of the month, the PLT batch list as at previous midnight will serve as the base PLT status. It will have details of all PLT batches that are already pumped and are in transit to the Refinery.

One base PLT batch list consists of details like Batch number, Batch name, Blending Ratio, Crude grade, ratio, Density, BS&W, viscosity, Quantity and expected date of delivery to refinery. In addition to this, any other desirable information can be added. Now there is one thumb rule that needs to be followed - *The total quantity of crude in pipeline is fixed. So, whenever a batch is created to push certain quantity in pipeline, the same quantity of crude shall be taken out of the pipeline at the receiving end, so that the total quantity in pipeline remains constant.* One special logic is designed to take care of this thing.

In this module, an array is created for daily PLT batch data input. The number of rows in this array may be anything depending on the maximum PLT batch list that is possible in a worst-case scenario on a particular day, considering good cushion for errors (like wrong grade pumping, emergency grade change, etc.).

Whenever a batch is created in module #3, it is copied and pasted in respective date wise array in module #4 in First in Last Out Sequence. Now, one PLT batch is uniquely defined by two parameters – Batch Name and Blending Ratio. So, when a new PLT batch is added, the system checks the existing PLT batch list (Previous day midnight) if that batch is present or not. If yes, then only quantity is updated, remaining all things (Batch number, Batch name, Blending Ratio, Crude grade, ratio, Density, BS&W, viscosity) remain same. If No, then the new batch is inserted on top of the previous day's PLT batch list and deducts the quantity of new batch from the last batch of the updated PLT batch list. And there may be a case where the value after deduction comes out to be negative in the bottom most PLT. In that case the negative value is adjusted in the second last batch of the updated PLT batch list and so on, till the total sum of updated batch list equals the fixed constant pipeline quantity.

Figure 6—A screenshot of background working in Module #3 of COIMP.

Figure 7—A screenshot of a part of Module #4 of COIMP. Data entry only in colored cells.

6510 cells are formulated with complex formulas for 10 PLT batch entry per day for 31 days.

In this module, all the data recorded in Module #1 – tank dips, temperatures, density and BS&W is used to calculate inventory in each tank in KL, MT, gross, Net, etc. This module contains calculation blocks for gross observed volume, floating roof weight correction, corrected gross observed volume, volume correction factor, weight correction factor, etc. All this is done in three different blocks, one for the current day, one for the last day and one for intermediate changes (R/D, D/R cases), if any. Then final stocks are calculated using simple addition or subtraction of these 3 blocks. 1362 cells are formulated in this module for one day calculation for 6 tanks.

The daily report in this module contains all the information about the crude oil movements done on a particular day. It contains the complete oil stock report – tank status, crude grades, crude ratios, opening tank dips, opening stocks, closing tank dips, closing stocks, crude receipt quantity, PLT quantity, pumpable stocks, ullage and dead stocks. Apart from this, it also contains the complete PLT batch list status for a

particular day, depicted in figures and graphical representation also. All this data is linked to the 5 modules described above and is updated by changing the date only. 313 cells are formulated in this module.

Everything else

The work described above is what I have been able to make a reality. Each of the 6 modules has the capability of accommodating customized changes. Complete structure of inputs and outputs can be restructured as per user requirements. Once we have all the data inputs captured correctly, rest only depends on the needs of user – any kind of report or pictorial or graphical representation of data for different use cases can be obtained from this program.

One module can have tank wise crude receipt detailing which vessel and which crudes entered and left the tank. Another module we can have motor tanker wise bifurcation of crude receipts. These are just examples. N number of data combination outputs are achievable with a high degree of ease. Only limited to the best of the imagination of a User.

A model for future prediction can also be created using COIMP. Say a motor tanker having 130 TKL Urals grade is to be unloaded in this terminal. COIMP can be customized based on user desirable outputs, to calculate and tell us how much crude is to be topped up and in which tank. Desirable outputs could be either grade/ratio, density, viscosity, or all of them.

Another model for crude receiving terminal (buffer) which received crude oil from COT via PLT and has the option to mix the PLT batches before feeding to CDU, will work in a similar manner. Twin stream blended Pipeline transfer batches can be received in different tanks based on a forecast given by COIMP based on desirable user inputs.

The best of my imagination is – user only needs to input the desirable production targets of a refinery and the system itself calculates everything upstream, till procurement needs. Extremely complex, but not impossible.

	Total stock (KL) @ ambient temp	Net Stock @ 15 °C						Gross @ 15 °C				Description	GROSS DIP	WATER DIP	TEMPERATURE: Deg. C	TOTAL OBSERVED VOLUME	FREE WATER	GROSS OBSERVED VOLUME	ROOF WEIGHT	OBSERVED DENSITY	
		Total stock			Total pumpable stock			Total Ullage													Dead Stock
		KL	MT	TMT	KL	MT	TMT	KL	MT	TMT	TMT										
1A	57123	56041	48828	48.8	47613	41486	41.5	0	0	0.0	7.3	1A	1584.9	0.00	37.50	57531.870	0.000	57531.870	350.133	0.8554	
1B	52595	51675	44322	44.3	43209	37060	37.1	4092	3510	3.5	7.3	1B	1460.4	0.00	34.50	53012.520	0.000	53012.520	352.241	0.8437	
1C	0	0	0	0.0	#VALUE!	#####	#####	#VALUE!	#####	#####	#VALUE!	1C	0.0	0.00	0.00	0.000	0.000	0.000	205.530	#VALUE!	
1D	57111	56095	47799	47.8	47537	40507	40.5	0	0	0.0	7.3	1D	1584.9	0.00	35.40	57531.870	0.000	57531.870	352.241	0.8374	
1E	57278	56419	49147	49.1	47883	41711	41.7	0	0	0.0	7.4	1E	1589.2	0.00	32.30	57687.960	0.000	57687.960	352.241	0.8589	
1F	53185	52363	43535	43.5	44036	36612	36.6	3503	2913	2.9	6.9	1F	1477.0	0.00	32.10	53615.100	0.000	53615.100	352.241	0.8188	
540039 531047 454866 454.7 #VALUE! ##### #VALUE! ##### #VALUE! ##### #VALUE! #####																					
122 DSR Date column number																					
	Total stock (KL)	Net Stock @ 15 °C						Gross @ 15 °C				Description	GROSS DIP	WATER DIP	TEMPERATURE: Deg. C	TOTAL OBSERVED VOLUME	FREE WATER	GROSS OBSERVED VOLUME	ROOF WEIGHT	OBSERVED DENSITY	
		Total stock			Total pumpable stock			Total Ullage													Dead Stock
		KL	MT	TMT	KL	MT	TMT	KL	MT	TMT	TMT										
1A	26690	26289	22905	22.9	17826	15532	15.5	29644	25829	25.8	7.4	1A	746.5	0.00	32.70	27097.950	0.000	27097.950	350.133	0.8588	
1B	52602	51666	44314	44.3	43203	37055	37.1	4084	3503	3.5	7.3	1B	1460.6	0.00	34.90	53019.780	0.000	53019.780	352.241	0.8435	
1C	#VALUE!	#VALUE!	#####	#####	#VALUE!	#####	#####	#VALUE!	#####	#####	#VALUE!	1C	0.0	0.00	0.00	0.000	0.000	0.000	205.530	#VALUE!	
1D	57118	56091	47795	47.8	47535	40504	40.5	0	0	0.0	7.3	1D	1585.1	0.00	35.60	57539.130	0.000	57539.130	352.241	0.8372	
1E	57274	56399	48129	48.1	47865	41695	41.7	0	0	0.0	7.4	1E	1589.1	0.00	32.70	57684.330	0.000	57684.330	352.241	0.8586	
1F	41704	41169	34228	34.2	32818	27285	27.3	14850	12346	12.3	6.9	1F	1160.7	0.00	29.20	42133.410	0.000	42133.410	352.241	0.8209	
#VALUE! #VALUE! ##### #VALUE! ##### #VALUE! ##### #VALUE! ##### #VALUE! #####																					
Prev. Day																					
	Total stock (KL)	Net Stock @ 15 °C						Gross @ 15 °C				Description	GROSS DIP	WATER DIP	TEMPERATURE: Deg. C	TOTAL OBSERVED VOLUME	FREE WATER	GROSS OBSERVED VOLUME	ROOF WEIGHT	OBSERVED DENSITY	
		Total stock			Total pumpable stock			Total Ullage													Dead Stock
		KL	MT	TMT	KL	MT	TMT	KL	MT	TMT	TMT										
1A	#VALUE!	#VALUE!	#####	#####	#VALUE!	#####	#####	#VALUE!	#####	#####	#VALUE!	1A	0.0	0.00		0.000	0.000	0.000	350.133	#VALUE!	
1B	#VALUE!	#VALUE!	#####	#####	#VALUE!	#####	#####	#VALUE!	#####	#####	#VALUE!	1B	0.0	0.00		0.000	0.000	0.000	352.241	#VALUE!	
1C	#VALUE!	#VALUE!	#####	#####	#VALUE!	#####	#####	#VALUE!	#####	#####	#VALUE!	1C	0.0	0.00		0.000	0.000	0.000	205.530	#VALUE!	
1D	#VALUE!	#VALUE!	#####	#####	#VALUE!	#####	#####	#VALUE!	#####	#####	#VALUE!	1D	0.0	0.00		0.000	0.000	0.000	352.241	#VALUE!	
1E	#VALUE!	#VALUE!	#####	#####	#VALUE!	#####	#####	#VALUE!	#####	#####	#VALUE!	1E	0.0	0.00		0.000	0.000	0.000	352.241	#VALUE!	
1F	#VALUE!	#VALUE!	#####	#####	#VALUE!	#####	#####	#VALUE!	#####	#####	#VALUE!	1F	0.0	0.00		0.000	0.000	0.000	352.241	#VALUE!	
#VALUE! #VALUE! ##### #VALUE! ##### #VALUE! ##### #VALUE! ##### #VALUE! #####																					
R/D pattern																					

Figure 8—A screenshot of a part of Module #5 of COIMP.

Table 1—Calculations used for one tank in module 5. Entry by user can be manual or with some integrated software like exaquantum, IP21, etc.

Sl. No.	Description	UOM	Data reference	Remarks
1	Tank dip	CM	Entry by User or automated system fetch.	
2	Tank Water cut	CM	Entry by User or automated system fetch.	
3	Observed Temperature	Deg. C	Entry by User or automated system fetch.	
4	Total Observed Volume (TOV) at observed Temp.	KL	Referred from Tank calibration chart	Volume of cargo at observed temperature.
5	Observed volume of Free Water	KL	Referred from Tank calibration chart	
6	Gross observed volume (GOV)	KL	TOV - GOV	
7	Roof Weight	MT	Referred from Tank calibration chart	
8	Observed density	gm/cc	Entry by User or automated system fetch.	Density at observed temperature.
9	Floating roof correction Volume (FRCV)	KL	Roof weight in MT / observed density in gm/cc	This correction is applied in case of floating roof tanks. It gives us the volume of cargo displaced by the weight of floating roof.
10	Corrected Gross observed volume (CGOV)	KL	GOV – FRCV	
11	Density @15Deg C	gm/cc	Entry by User or automated system fetch.	Measured using ASTM D5002. This is density in vacuum.
12	Volume Correction factor (VCF)	-	$\exp\{-\alpha_T \Delta T [1 + 0.8\alpha_T (\Delta T + \delta_T)]\}$	VCF is a factor used to convert crude volumes at different temperatures. Say, from standard 15 Deg C to observed temp. and vice-a-versa.
13	Gross Standard Volume (GSV)	KL	CGOV x VCF	
14	Weight correction factor (WCF)	gm/cc	Density @15 – 0.0011	WCF is used to convert weight of crude from vacuum to air and vice-a-versa.
15	Gross Metric Tonnes	MT	GSV x WCF	Mass in air.
16	BS&W	%	Entry by User or automated system fetch.	
17	BS&W Volume	KL	GSV x BS&W%	
18	Net Standard Volume (NSV)	KL	GSV – BS&W Volume	
19	BS&W Mass	MT	Gross Metric Tonnes x BS&W%	
20	Net Metric Tonnes	MT	Gross Metric Tonnes - BS&W Mass	
21	Dead Stock Volume (DSV)	KL	Referred from Tank calibration chart	Inventory that cannot be pumped.
22	Dead Stock with roof weight correction	KL	DSV - FRCV	

Use of Artificial Intelligence in making COIMP

Artificial Intelligence is the buzz word in the ever-evolving industrial landscape of earth. AI is a tool that brings skills to people, learning which could take a lifetime. In a world without AI, this program (COIMP) would have required one to master MS Excel, spending time learning formulas and features in different versions of it. Role of AI has been pivotal in gaining expertise in MS Excel by saving all that time which one normally requires without AI to do the task. It is more like revealing the secrets of software like an infiltrator,

which ultimately resulted in improving my understanding of MS Excel's capabilities. No doubt AI made me a 100% Excel efficient guy, still, there is some skill required to get AI do the job for you.

In case of COIMP, AI did not produce the program in one single prompt. In fact, AI wasn't even aware of what I am doing with all those questions about MS EXCEL. Brick by brick, step by step, I asked AI to solve problems that I encountered while designing COIMP. These problems were with data linking, references and calculations using different formulas in MS Excel. At times, more complex calculation or search blocks (which are not available in MS Excel) were formulated using the basic MS Excel formula database. Again, designed by self, simplified by AI. So, my queries for AI would be as simple as how to get a particular logic work, or which formula and in what syntax can be used to minimize or optimize a complex calculation. Sometimes I had to break a complex logic into baby steps. Solving each baby step and structuring together would make the logic work.

Use of AI in Module #2: A simple example is from Module #2, where my first task was to read and separate the individual grades of crude written in one cell as ARH:ARL:BHY:URL. So, I asked AI, how to do it? AI suggested to use the formula "TEXTSPLIT" and the version of MS Excel in which it is available. Then Same with ratios. Split 20:30:10:40 into different cells. Next add an array for new grades and quantities, for which you don't need AI. After that resultant block is created where based on added quantities, a tank's crude grade or ratio is updated using simple and complex formulas, both. Complex formulas used multiple simple formulas like MATCH, MAX, MIN, IF, COLUMN, VALUE, etc. I would ask AI different questions to set up a perfect and most optimum combination of simple formulas. So on and so forth, new problems popped up. One of them was that the sum of all grade ratios in some cases was not coming out to be 100%. Then I set up a complex 100% check block to ensure that all tank grade ratio percentages are exactly 100%. All set, now merge all the results and display in 2 cells, one for updated grade and one for updated ratio post crude receipt. AI suggested to use "TEXTJOIN" for this. This is just a part of the problem-solving technique used by me in line with MS Excel and AI for making different modules of COIMP. What sets me apart from AI or MS Excel is the structuring and logic designing of the program which was completely done by me.

Use of AI in Module #3: Here is the best use case of AI in making one specific block which is used in creating PLT batch Crude Grades and Ratios. Refer Fig. 5, where users only need to enter the batch name and blending ratio. This specific block fetches the crude grades and ratios from the individual tanks from Module #2. Pro rates the ratios in line with blending ratios. Then, it searches for crude grades one tank in another and vice-a-versa. If both tanks have same grades, pro-rated ratios are added. If they have unique grades, then only pro-rated ratios are considered in final PLT grades and ratios. This special logic block is shown in first 15 rows in Fig. 6. In this, even the empty cells have formulas to nullify errors generated during the operation of this block. Each of the above-described steps generated some error. These errors were solved using AI. Like this only, AI is used in other modules of COIMP.

Exciting, isn't it? Because one doesn't need to remember things with MS Excel. One need not search detailed forums on internet trying to solve just one simple problem. Or refer to old worksheets for specific logic that were prepared long ago in Excel. This may be true for most softwares. Saving Time is the first exciting thing. With growing competition, users are getting better versions of AI on the internet. There is competition to offer the best features in freeware AI programs to lure customers into buying AI softwares. Freeware AI alone has huge capabilities. COIMP was compiled using freeware AI only. Not a single penny spent on AI. COIMP gives good competition to its licensed counterpart softwares, which are used for the same task by various industry players. Plus, it is highly customizable. This is just one industrial use case. COIMP has all the scope and margins of improvement to actually replace its licensed counterpart software, which can result in capital savings without compromising reliability of the software and without revealing any company sensitive info to the AI servers.

COIMP has gone through months of testing in different operating scenarios in our crude oil terminal. There is a fine accuracy in the results generated by it. Although, there is more scope of improvement, which, as I earlier mentioned, is only limited to the best of a User's imagination.

Tank	Tank Crude Mix (%)			Opening Level (cm)	Opening Stock (TMT)	Receipts Qty (TMT)	Daily Pumping Qty (TMT)	Closing Level (cm)	Closing Stock (TMT)	Pumpable Stock (TMT)	Ullage (TMT)	Dead Stock (TMT)
	R/D/S/M*	Grade	Ratio									
1A	D	ARL:ALS:URL	89:5:6	860.1	26.0	0.0	14.0	401.6	12.0	4.7	36.0	7.2
1B	S	ARL:MUR:URL	79:3:18	1249.4	38.0	0.0	0.0	1249.4	38.0	30.7	10.0	7.3
1C	M		-	-	-	-	-	-	-	-	-	-
1D	S	ARL:SKF:URL	76:11:13	1582.1	47.8	0.0	0.0	1582.5	47.8	40.5	0.0	7.3
1E	D	ARL:ALS:URL	2:6:92	1471.8	45.5	0.0	9.1	1180.5	36.4	29.0	12.3	7.4
1F	S	ARL:ARM:MUR	12:22:66	1475.1	44.1	0.0	0.0	1475.2	44.1	37.1	3.0	7.0
*R-Receiving/ D-Dispatch/ S-StandBy/ M-Maintenance					327.7	32.2	23.1		336.9	253.2	194.1	83.5

1A:1E

1A:1B

1B:1C

1C:1D

1D:1E

1E:1F

1F:1A

1A:1B

COT END

Refinery END

VBPL PIPELINE											Remarks :			
Batch No.	Tank Name	Blending Ratio %	Qty (KL)	Qty (MT)	Grade	Ratio	Density	BS&W %	FS (PPM)	Viscosity	Exp. Arrival	Pumping Qty of previous day as per ATG system	27218	KL
2740	1A:1E	60:40	40667	35051	ARL:ALS:URL	55:5:40	0.8619	0.070	443	11.0	11-08-2025 03:49	Pumping Qty of previous day as per Flow Meter	27149	KL
2739	1A:1B	65:35	50073	43253	ARL:URL:ALS	37:61:2	0.8638	0.083	394	12.5	09-08-2025 07:39	Receipt Qty of previous day as per ATG system	37812	KL
2738	1B:1C	60:40	24683	21378	ARL:UPP:URL	23:16:61	0.8661	0.082	318	12.5	08-08-2025 09:54	DRA Consumption	173.2	Litres
2737	1C:1D	70:30	66106	57327	ARL:UPP:URL	24:12:64	0.8672	0.077	231	12.7	05-08-2025 23:36	** VBPL PLT in progress from Tanks-1I & 1A.		
2736	1D:1E	60:40	8190	7070	ARL:ARM:BHY:MUR:UPP:URL	23:20:2:17:16:22	0.8633	0.052	418	10.2	05-08-2025 16:23	** Tank-1C: Major M&I jobs are in progress.		
2735	1E:1F	60:40	28822	25130	ARL:ARM:UPP:URL	18:38:16:28	0.8719	0.112	292	11.3	04-08-2025 14:58			
2734	1F:1A	60:40	6628	5800	ARL:ALS:ARM:MUR:URL	6:15:38:1:40	0.8751	0.130	377	13.0	04-08-2025 09:07			
2733	1A:1B	60:40	37573	32553	ARL:URL:MUR:ALS	36:47:2:15	0.8664	0.100	501	12.1	03-08-2025 00:00			

Figure 9—A screenshot of Module #6 of COIMP.